

UNIVERSITY OF BRISTOL

MSC BIOINFORMATICS

**Effect of Forest Age on Machine Learning Generalisation of
Evapotranspiration Using FLUXNET Data**

PROJECT PLAN

Author

Archie BENN

Supervisor

Professor Martin DE KAUWE

ABSTRACT (The abstract has a maximum of 200 words)

Key words: Machine Learning, FLUXNET, Evapotranspiration.

Word count: 3000

DECLARATION

I declare that the work in this dissertation was carried out in accordance with the requirements of the University's Regulations and Code of Practice for Taught Programmes and that it has not been submitted for any other academic award. Except where indicated by specific reference in the text, this work is my own work. Work done in collaboration with, or with the assistance of others, is indicated as such. I have identified all material in this dissertation which is not my own work through appropriate referencing and acknowledgement. Where I have quoted or otherwise incorporated material which is the work of others, I have included the source in the references. Any views expressed in the dissertation, other than referenced material, are those of the author.

Archie BENN

May 2, 2026

CONTENTS

Literature Review (3000 words)	4
Introduction ~800 Words	4
Evapotranspiration	4
Canopy Cover and Vegetation Indices (CHANGE)	4
Climate Change and Forest Age	5
Evapotranspiration Prediction ~1200 Words	
<i>(What don't we know and what are the issues?)</i>	5
Contemporary ET Prediction Methods	5
FLUXNET	5
Machine Learning Models for ET Prediction	5
Machine Learning Applicability ~600 Words	
<i>(What is the case for using ML here?)</i>	5
Why use Machine Learning?	5
Machine Learning Generalisation?	5
Project Aims and Objectives (500 words)	6
Aims	6
Project Plan (500 words)	7
Data Acquisition	7
Site Selection	7
Analysis	7
Feature Engineering	7
ML Model Setup and Training	7
ML Model Evaluation	7
ML Model Predictions	7
Tools	7
Interpretation	7
Gantt Chart	7
Risk Register (~ 250 words)	8
Figures and Tables	9

LITERATURE REVIEW (3000 WORDS)

INTRODUCTION ~800 WORDS

Evapotranspiration

Terrestrial evapotranspiration (ET) is a key component of the global hydrological cycle describing the transition of water from liquid to gas at Earth's surface and further transfer to the atmosphere, and is a crucial component of the energy and water exchange between terrestrial ecosystems and the climate system. The term emphasises the combination of water vapourisation flux from (1) abiotic evaporation from soil and plant canopy surfaces, and (2) biotic transpiration through stomata to the atmosphere [Katul et al., 2012]. ET can be generally described through:

$$ET = T + E_{Soil} + E_{Canopy} \quad (1)$$

where T refers to transpiration through plant stomata, E_{soil} to evaporation from the soil surface, and E_{Canopy} to evaporation from plant canopies. The contribution of these variables to overall ET is not equal, with Lian et al. [2018] estimating a global T/ET ratio of 0.62 ± 0.06 , indicating plant transpiration as the leading component of ET worldwide. The partitioning of ET into E ($E_{soil} + E_{Canopy}$) and T within ecosystems is associated with "water use efficiency" and has use in irrigation management, where T is linked with plant productivity, while E does not directly contribute to production [Kool et al., 2014]. This in turn likely explains the elevated E/ET ratios in managed croplands compared to natural ecosystems [Yang et al., 2023]. Accurate partitioning estimates are also important in aiding carbon cycle projections given the coupled effect of transpiration and land-atmosphere carbon dioxide exchanges through stomatal conductance.

Canopy Cover and Vegetation Indices (CHANGE)

ET is governed by many environmental factors such as solar radiation, leaf area index (LAI), air temperature, vapour pressure deficit (VPD), soil moisture, and wind speed [Yang et al., 2023]. Of these, Lian et al. [2018] identified LAI as the primary driver of T/ET spatial patterns, and Kool et al. [2014] suggest that in systems with full canopy cover (and therefore high LAI) ET is often assumed to be similar enough to T to permit correlation of ET with biomass. However, this assumption may break down in systems with multi-layer canopies such as forests despite high LAI. In these systems, the 2-dimensional measure of LAI will not capture the vertical canopy complexity which drives rainfall interception and subsequent E_{Canopy} , implying that canopy cover indices may struggle to explain T/ET ratios in more complex vertical environments. This is reflected by Zhou et al. [2016] who report higher coefficients of determination between enhanced vegetation index (EVI) and T/ET in corn and soybean sites (0.84 and 0.82) with simple canopies, compared to evergreen needle leaf forests (0.37).

[link to how this means there is more to it than assuming high canopy cover = more T = more biomass and ow forests are complex systems making this hard. and how age affects canopies etc.]

Climate Change and Forest Age

EVAPOTRANSPIRATION PREDICTION ~1200 WORDS

(What don't we know and what are the issues?)

Contemporary ET Prediction Methods

FLUXNET

Machine Learning Models for ET Prediction

MACHINE LEARNING APPLICABILITY ~600 WORDS

(What is the case for using ML here?)

Why use Machine Learning?

Machine Learning Generalisation?

PROJECT AIMS AND OBJECTIVES (500 WORDS)

Aims

- Overview of how this project will answer the question/area proposed at end of introduction.

PROJECT PLAN (500 WORDS)

DATA ACQUISITION

SITE SELECTION

ANALYSIS

Feature Engineering

ML Model Setup and Training

ML Model Evaluation

ML Model Predictions

TOOLS

INTERPRETATION

GANTT CHART

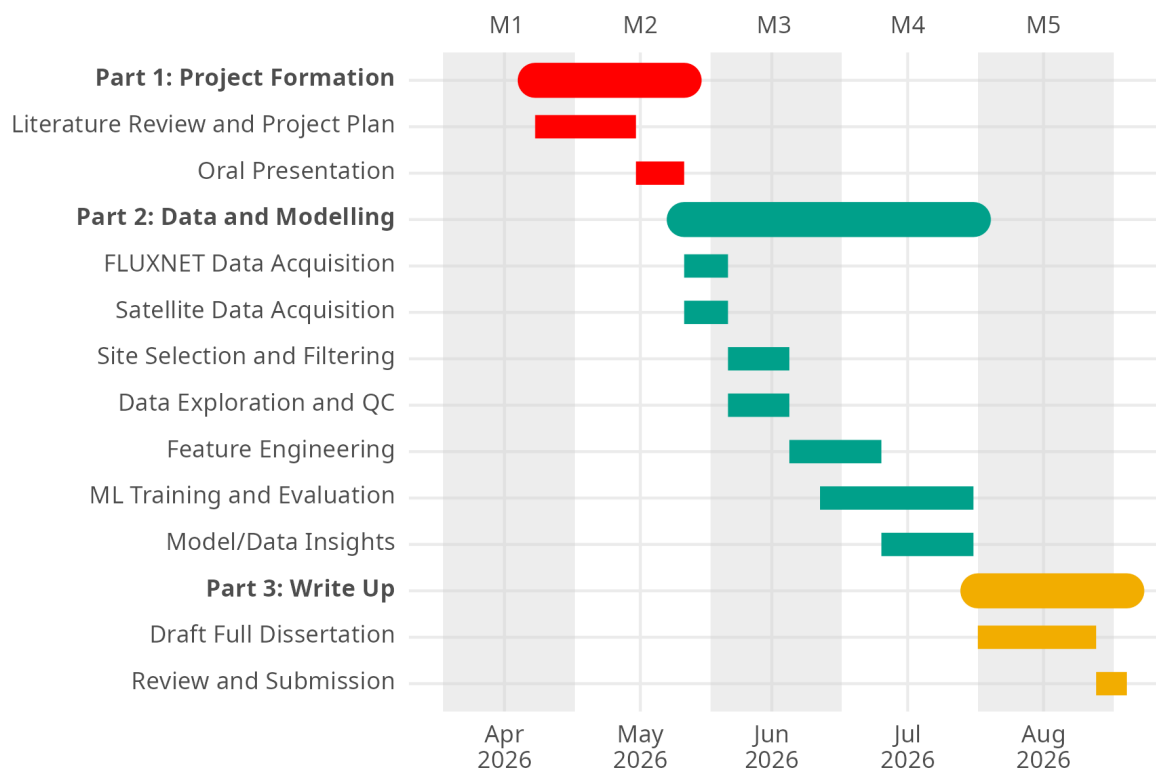


Figure 1: A proposed Gantt Chart for the duration of this project.

RISK REGISTER (~ 250 WORDS)

REFERENCES

- Katul, G. G., Oren, R., Manzoni, S., Higgins, C., and Parlange, M. B. (2012). Evapotranspiration: A process driving mass transport and energy exchange in the soil-plant-atmosphere-climate system. *Reviews of Geophysics*, 50(3).
- Kool, D., Agam, N., Lazarovitch, N., Heitman, J., Sauer, T., and Ben-Gal, A. (2014). A review of approaches for evapotranspiration partitioning. *Agricultural and forest meteorology*, 184:56–70.
- Lian, X., Piao, S., Huntingford, C., Li, Y., Zeng, Z., Wang, X., Ciais, P., McVicar, T. R., Peng, S., Ottlé, C., et al. (2018). Partitioning global land evapotranspiration using cmip5 models constrained by observations. *Nature Climate Change*, 8(7):640–646.
- Yang, Y., Roderick, M. L., Guo, H., Miralles, D. G., Zhang, L., Fatichi, S., Luo, X., Zhang, Y., McVicar, T. R., Tu, Z., et al. (2023). Evapotranspiration on a greening earth. *Nature Reviews Earth & Environment*, 4(9):626–641.
- Zhou, S., Yu, B., Zhang, Y., Huang, Y., and Wang, G. (2016). Partitioning evapotranspiration based on the concept of underlying water use efficiency. *Water Resources Research*, 52(2):1160–1175.

FIGURES AND TABLES

Method	Advantages	Disadvantages
--------	------------	---------------